

Required Practical Activities (PAGs)

OCR A GCSE Biology (J247) Gateway — Practical Activity Groups

OCR A GCSE Biology (Gateway) requires you to complete at least eight practical activities, grouped into eight Practical Activity Groups (PAGs): **B1** Microscopy, **B2** Testing for biological molecules, **B3** Sampling techniques, **B4** Rates of enzyme-controlled reactions, **B5** Photosynthesis, **B6** Physiology, responses and respiration, **B7** Microbiological techniques and **B8** Transport in and out of cells. Each PAG is defined by the apparatus and techniques it covers, so your school may use different activities; the practicals below are typical examples for each group. Examiners will test your understanding of them in the written papers. Each practical shows the method, key apparatus, expected results and how to handle the data.

Exam Tip: When asked about a practical in the exam, always: (1) state the independent and dependent variables and at least 2 control variables; (2) describe how the dependent variable is measured; (3) explain how repeats and means improve precision; (4) suggest one source of error and how to minimise it.

PAG B1 — Microscopy

Aim: Use a light microscope to observe and draw plant cells; calculate magnification.

Apparatus: Light microscope, slide, coverslip, onion bulb, forceps, iodine (or methylene blue) stain, mounted needle, tile, scalpel, ruler.

Method:

1. Peel off a thin layer of onion epidermis using forceps.
2. Place flat on a slide; add a drop of iodine stain.
3. Lower a coverslip carefully using a mounted needle (avoid air bubbles).
4. Place slide on stage; secure with clips.

5. Focus at lowest magnification first: watching from the side, use the coarse focus to bring the stage close to the objective, then move the stage AWAY while looking through the eyepiece until the image is sharp.
6. Switch to higher magnification — only use FINE focus.
7. Draw a clear diagram, including a scale bar, label and magnification.

Calculations:

- Total magnification = eyepiece $\times$ objective.
- Magnification = image size $\div$ real size; convert units (1 mm = 1000 μ m).

Hazards: Sharp scalpel, fragile slides — handle with care.

PAG B2 — Testing for Biological Molecules (food tests)

Aim: Identify the presence of starch, sugars, lipids and proteins in food samples.

Molecule	Reagent	Positive result
Starch	Iodine solution	Orange-brown $\rightarrow$ blue-black
Reducing sugars (e.g. glucose)	Benedict's reagent + heat (80 °C)	Blue $\rightarrow$ green $\rightarrow$ yellow $\rightarrow$ orange $\rightarrow$ brick red (more sugar = redder)
Proteins	Biuret reagent	Blue $\rightarrow$ purple/violet
Lipids	Ethanol + water (emulsion test)	Cloudy white emulsion

Hazards: Hot water bath; iodine stains skin; ethanol is flammable.

PAG B3 — Sampling Techniques: Random Quadrat Sampling

Aim: Estimate the population of a plant species in a habitat.

Method:

1. Mark out the habitat area; measure it (e.g. $10\text{ m} \times 10\text{ m} = 100\text{ m}^2$).
2. Generate random coordinates using a random number generator.
3. Place a $0.5\text{ m} \times 0.5\text{ m}$ quadrat at each random position. Count target plants inside.
4. Repeat 10–20 times.
5. Calculate the mean number per quadrat.
6. Estimate total population = mean $\times$ (total area $\div$ area of quadrat).

Calculation example: If mean = 6 plants per 0.25 m^2 quadrat in a 100 m^2 field: total = $6 \times (100 \div 0.25) = 2400$.

Variables: IV = location (random); DV = number of plants. Repeats reduce the impact of patchy distribution.

PAG B3 — Sampling Techniques: Transect Investigation

Aim: Investigate how an environmental gradient affects the distribution of species (e.g. distance from a hedge, light vs shade).

Method:

1. Stretch a tape measure (transect) from the start of the gradient (e.g. base of hedge) to the end (e.g. middle of field).
2. At regular intervals (e.g. every 1 m), place a quadrat next to the tape.
3. Count target species OR estimate % cover for each species in each quadrat.
4. Also record abiotic conditions (e.g. light intensity, soil moisture) at each quadrat.
5. Plot graphs of species abundance (y) against distance along transect (x).

Result interpretation: Look for trends — species declining/increasing with distance; correlate with abiotic data.

PAG B4 — Rates of Enzyme-Controlled Reactions: Effect of pH on Amylase

Aim: Investigate how pH affects the rate at which amylase digests starch.

Method:

1. Set up buffer solutions at pH 3, 5, 7, 9, 11.
2. Add a drop of iodine to each well of a spotting tile (one row per pH).
3. Mix equal volumes of starch solution, amylase and the chosen buffer in a test tube; place in 37 °C water bath.
4. Every 30 seconds, take a drop of mixture and add it to a fresh iodine well.
5. Time how long until iodine no longer turns blue-black (no starch left). Record.
6. Repeat 3 times at each pH and calculate mean time.

Variables: IV = pH; DV = time (s) for starch to be fully digested. Control: temperature, volume/concentration of starch and amylase, time intervals.

Calculations: Rate = $1 \div \text{time (s)}$; plot rate vs pH — peak gives optimum pH (~7 for amylase).

Hazards: Buffer solutions can irritate; iodine stains.

PAG B5 — Photosynthesis: Effect of Light Intensity on Pondweed

Aim: Investigate how light intensity affects the rate of photosynthesis in Cabomba pondweed.

Method:

1. Cut a sprig of Cabomba and place it (cut end up) in a beaker of water with a small amount of NaHCO_3 to provide CO_2 .
2. Place a lamp at distance 10 cm from the beaker. Wait 5 minutes for the plant to acclimatise.
3. Count bubbles released from the cut stem in 1 minute. Repeat 3 times; calculate mean.
4. Move lamp to 20 cm, 30 cm, 40 cm, 50 cm and repeat counts.
5. Calculate light intensity $\propto 1/d^2$; plot rate (y) vs $1/d^2$ (x).

Variables: IV = light intensity (controlled by lamp distance); DV = bubble count per minute.
Control: temperature (heat sink between lamp and plant), volume of water and NaHCO_3 , type/length of pondweed.

Hazards: Hot lamp — handle carefully; water near electricity — keep dry.

PAG B6 — Physiology, Responses and Respiration: Rate of Respiration in Germinating Peas

Aim: Measure the rate of oxygen consumption by germinating peas using a respirometer.

Method:

1. Place germinating peas (kept moist for 2 days) into a sealed boiling tube with soda lime (absorbs CO_2).
2. Connect the tube to a calibrated capillary tube containing a coloured liquid drop.
3. Place in a water bath at constant temperature.
4. Allow apparatus to equilibrate for 5 min, then mark the starting position.
5. Measure how far the liquid moves in 5 minutes — this represents the volume of O_2 used.
6. Calculate rate of respiration = distance / time (or use cross-section to calculate volume).
7. Repeat with a control (boiled, dead peas) — should show no movement.

Variables: IV = type/age of peas (or temperature); DV = rate of O_2 consumption.

Hazards: Soda lime is corrosive — wear goggles.

PAG B6 — Physiology, Responses and Respiration: Reaction Time (ruler-drop test)

Aim: Measure reaction time and investigate the effect of a variable (e.g. caffeine, time of day, music).

Method:

1. Person A holds a 30 cm ruler vertically just above the open hand of person B.
2. Without warning, Person A drops the ruler.
3. Person B catches the ruler as quickly as possible — record the distance the ruler fell (cm).
4. Convert distance to time using $t = \sqrt{(2d/g)}$, where $g = 9.81 \text{ m/s}^2$. Or use a calibrated reaction-time chart.
5. Repeat 5 times and calculate mean.
6. Change ONE variable (e.g. give a small caffeinated drink and wait 20 min) and repeat.

Variables: IV = condition tested; DV = reaction time. Control: same person, same hand, same ruler, same start position, same time of day.

Calculation: e.g. $d = 0.20 \text{ m} \rightarrow t = \sqrt{(2 \times 0.20 \div 9.81)} \approx 0.20 \text{ s}$.

PAG B6 — Physiology, Responses and Respiration: Plant Responses (phototropism in cress seedlings)

Aim: Investigate how light direction affects the growth of plant shoots.

Method:

1. Sow cress seeds on damp cotton wool in three Petri dishes.
2. Dish 1: keep in light from above (control).
3. Dish 2: place in a closed box with a single hole on one side (light from one direction only).
4. Dish 3: keep in complete darkness (control).
5. Water all dishes equally and leave for 5–7 days at room temperature.
6. Observe and record the direction of growth.

Expected results: Dish 1 grows straight up. Dish 2 bends towards the light. Dish 3 still grows upwards (shoots respond to gravity) but the seedlings are etiolated: tall, thin, pale and yellow because no chlorophyll is made.

Explanation: Auxin produced at the shoot tip moves to the shaded side, causing more elongation there → shoot bends towards the light.

PAG B6 — Physiology, Responses and Respiration: Effect of Exercise on Pulse Rate

Aim: Investigate how exercise affects heart (pulse) rate and how quickly it returns to the resting rate.

Apparatus: Stopwatch (or a heart-rate monitor / data logger), step or bench, metronome (optional).

Method:

1. Sit quietly for 5 minutes. Find the pulse at the wrist (radial artery) or neck using two fingers — not the thumb, which has its own pulse.
2. Count beats for 15 seconds and multiply by 4 to get beats per minute. Repeat to get a mean resting rate.
3. Carry out a set exercise for a fixed time (e.g. step-ups at a steady pace for 2 minutes).
4. Immediately measure the pulse, then measure it again every minute until it returns to the resting rate. Record the recovery time.
5. Repeat on another occasion, or change the intensity/duration of exercise as the independent variable.

Variables: IV = intensity or duration of exercise; DV = pulse rate (bpm) and recovery time.

Control: same person, same type of exercise, same rest period beforehand, same body position when measuring.

Explanation: During exercise muscles respire faster, so more oxygen and glucose must be delivered and more CO₂ removed — heart rate and breathing rate rise. After exercise the rate stays high for a while to repay the oxygen debt (breaking down lactic acid). Fitter people usually have a lower resting rate and recover faster.

Hazards: Anyone with a medical condition (e.g. asthma, heart problems) should not take part; stop if feeling faint or dizzy.

PAG B7 — Microbiological Techniques: Effect of Antiseptics or Antibiotics on Bacterial Growth

Aim: Investigate the effectiveness of different antiseptics/antibiotics by measuring zones of inhibition.

Method (aseptic technique):

1. Sterilise the inoculation loop using a Bunsen burner flame.
2. Spread a lawn of bacteria onto the agar plate using a sterile spreader (work near a flame).
3. Soak paper discs in different antiseptic/antibiotic solutions; include a water-only disc as control.
4. Place the discs evenly on the agar (use sterile forceps).
5. Tape the lid (do not seal completely — to maintain aerobic conditions and prevent dangerous pathogens growing); incubate at 25 °C for 48 hours (in schools — pathogens grow at 37 °C).
6. Measure the diameter of clear "zones of inhibition" around each disc.
7. Calculate the area of inhibition using $A = \pi r^2$ (where r = radius).

Variables: IV = type/concentration of antibiotic; DV = area of inhibition (cm²). Control: same bacterium, same agar, same incubation conditions.

Hazards: Bacteria — even non-pathogens can grow rapidly. Always use aseptic technique. Dispose of plates by autoclaving.

PAG B8 — Transport In and Out of Cells: Osmosis in Potato Cylinders

Aim: Investigate how the concentration of sucrose solution affects mass change in potato tissue.

Method:

1. Use a cork borer to cut potato cylinders of equal diameter.
2. Cut cylinders to the same length (e.g. 3 cm) using a scalpel and ruler; blot dry with tissue.
3. Record the initial mass of each cylinder using a balance.
4. Place each cylinder in a different sucrose concentration: 0.0, 0.2, 0.4, 0.6, 0.8, 1.0 mol/dm³.
5. Leave at constant temperature for 30 minutes.
6. Remove, blot dry, and record the final mass.
7. Calculate % change in mass = $(\text{final} - \text{initial}) \div \text{initial} \times 100$.
8. Plot % change (y) vs sucrose concentration (x). The point where line crosses zero estimates the cell sap concentration.

Variables: IV = sucrose concentration; DV = % change in mass. Control: starting size of potato, temperature, time, volume of solution.

Hazards: Sharp cork borer/scalpel; cut on a tile.